Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam A · Form 2: Foundations

Multiple Choice Questions: Select the best answer. (1 point each)

1. A ball is launched horizontally at 3 m/s from a table 4.9 m high. Ignore drag. Find flight time. Use g = 9.8 m/s².

(A) 2 s

(B) 0.5 s

(C) 1 s

(D) -1 s

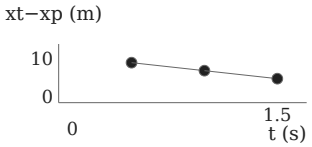
2. A projectile moves horizontally at +6 m/s while a target travels at +2 m/s. Track their horizontal separation during flight. t (s) -> xt-xp (m): (0.5, 10); (1, 8); (1.5, 6). As t (s) increases, what happens to xt-xp (m)?

(A) reversal midway

(B) decrease

(C) increase

(D) no change



3. Two balls are launched simultaneously from the same height horizontally. B has 2 times A’s horizontal velocity. Both land on the same ground without drag. Compare flight times and ranges.

(A) Flight times are equal; B’s range is 2 times A’s.

(B) B takes 2 times as long; ranges are equal.

(C) Both flight times and ranges are equal.

(D) Both B’s flight time and range are 2 times A’s.

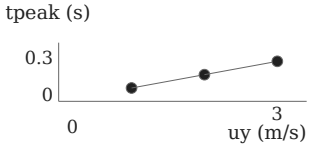
4. In gravity g=10 m/s² vary the upward component of launch velocity and time the ascent. uy (m/s) -> tpeak (s): (1, 0.1); (2, 0.2); (3, 0.3). What is the mean of the first two vertical readings?

(A) 0.21

(B) 0.09

(C) 0.15

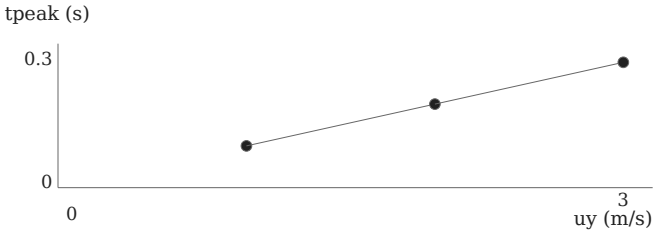
(D) 0.27



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. In gravity g=10 m/s² vary the upward component of launch velocity and time the ascent. uy (m/s) -> tpeak (s): (1, 0.1); (2, 0.2); (3, 0.3).

a. For the apex timing setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Plot the first and last measurements and describe the trend. **(3 pt)**

6. A map route starts at east coordinate 2 m and runs at 60° north of east. d (m) -> east (m): (1, 2.5); (2, 3); (3, 3.5).

a. Build the model for eastward projection from physics and predict both slope and intercept. **(2 pt)**

b. Plot the first and last measurements and describe the trend. **(2 pt)**

7. A projectile moves horizontally at +6 m/s while a target travels at +2 m/s. Track their horizontal separation during flight. t (s) -> xt-xp (m): (0.5, 10); (1, 8); (1.5, 6). Build the model for moving target from physics and predict both slope and intercept. **(3 pt)**